A SHARP ASYMPTOTIC THRESHOLD FOR TERMINAL EXTRAPOLATION OF GRADIENT DESCENT

RUI LI

ABSTRACT. We study the worst-case gradient norm at the terminal extrapolation $y_c = x_0 + c(x_N - x_0)$ of gradient descent on smooth convex functions. Let $c_{\mathrm{crit}}(N,h)$ be the supremum of the factors $c \geq 1$ for which the Huber bound $LD/(1+Nhc)$ is exact, where L is the gradient Lipschitz bound, D is the initial distance bound, and $h \in (0,1]$ is the normalized step size. For each fixed h,

$$c_{\mathrm{crit}}(N,h) - 1 \sim \frac{1}{hN\sqrt{\log N}}.$$

This disproves Conjecture 5.1 of Luner and Grimmer. The lower bound follows from an interpolation certificate and a tridiagonal energy estimate. For the upper bound, planar smooth convex examples and a one-dimensional quadratic give strict failure for every factor above the matching scale. The main result is formalized in Lean.

1. INTRODUCTION

After N gradient descent steps, terminal extrapolation replaces the final iterate by $y_c = x_0 + c(x_N - x_0)$, with $c \geq 1$. This output uses only the initial and final iterates and requires no additional gradient evaluations. Luner and Grimmer [2] established sharp worst-case function-value guarantees for terminal extrapolation and studied the corresponding gradient-norm problem numerically. In Conjecture 5.1 of the fourth arXiv version, they propose that the critical excess extrapolation factor is $\Theta(N^{-1})$ for each fixed normalized step size $h \in (0,1]$.

Let $c_{\mathrm{crit}}(N,h)$ denote the supremum of the factors for which the Huber gradient-norm bound is exact. We prove that, for each fixed $h \in (0,1]$,

$$c_{\mathrm{crit}}(N,h) - 1 \sim \frac{1}{hN\sqrt{\log N}}.$$

In particular, $N(c_{\mathrm{crit}}(N,h) - 1) \to 0$, disproving the conjecture.

The proof combines performance estimation [1] with smooth-convex interpolation [3]. The KKT multipliers of an auxiliary quadratic program determine a nonnegative directed flow and an interpolation certificate. A tail-sum transformation reduces the objective to a tridiagonal energy; comparison with a reciprocal model identifies its leading term. The resulting lower bound gives exactness below the critical scale. Above that scale, strictly feasible negative directions yield planar counterexamples by convex conjugacy, and a quadratic function covers the unbounded tail.

Sections 3–4 establish the certificate and lower bound. Sections 5–6 construct the counterexamples, and Section 7 completes the proof. Appendices A and B contain the flow construction and energy estimates; Appendix C gives the local transition law of the auxiliary program.

Date: October 15, 2026.

2020 *Mathematics Subject Classification.* 90C25, 90C30, 65K05.

Key words and phrases. Gradient descent, extrapolation, worst-case gradient norm, smooth convex interpolation, performance estimation, asymptotic analysis.

2. Setting and main theorem

For $L > 0$ and $d \geq 1$, let $\mathcal{F}_L(\mathbb{R}^d)$ be the class of convex differentiable functions $f : \mathbb{R}^d \to \mathbb{R}$ having an L-Lipschitz gradient and attaining their minimum. Let $x_\star$ be a minimizer with $\|x_0 - x_\star\| \leq D$, where $D > 0$. For $N \geq 1$, $0 < h \leq 1$, and $c \geq 1$, consider

$$x_{i+1} = x_i - \frac{h}{L}\nabla f(x_i) \quad (0 \leq i < N), \qquad y_c = x_0 + c(x_N - x_0). \tag{2.1}$$

Define the worst-case gradient norm by

$$P_{N,h}(c; L, D) = \sup\left\{ \|\nabla f(y_c)\| : \begin{array}{l} d \geq 1,\ f \in \mathcal{F}_L(\mathbb{R}^d), \\ x_\star \in \operatorname{argmin} f,\ x_0 \in \mathbb{R}^d,\ \|x_0 - x_\star\| \leq D \end{array} \right\}. \tag{2.2}$$

Define

$$c_{\mathrm{crit}}(N, h) = \sup\left\{ c \geq 1 :\ P_{N,h}(c; L, D) = \frac{LD}{1 + Nhc} \right\}. \tag{2.3}$$

Translation and scaling make c_{crit} independent of L, D. Throughout, log denotes the natural logarithm.

Theorem 2.1. *For every fixed $h \in (0, 1]$,*

$$\lim_{N\to\infty} hN\sqrt{\log N}\,(c_{\mathrm{crit}}(N, h) - 1) = 1. \tag{2.4}$$

More precisely, for every $h \in (0, 1]$ and $\varepsilon \in (0, 1)$, there exists $N_0 = N_0(h, \varepsilon) \geq 2$ such that, for all integers $N \geq N_0$ and all $L, D > 0$,

$$P_{N,h}(c; L, D) = \frac{LD}{1 + Nhc} \quad \text{for } 1 \leq c \leq 1 + \frac{1 - \varepsilon}{hN\sqrt{\log N}}, \tag{2.5}$$

whereas, writing $c_+ = 1 + (1 + \varepsilon)/(hN\sqrt{\log N})$,

$$P_{N,h}(c; L, D) > \frac{LD}{1 + Nhc} \qquad \text{for every } c \geq c_+. \tag{2.6}$$

For each $c \geq c_+$, the strict inequality is realized by a function in dimension at most two with $\|x_0 - x_\star\| = D$.

Corollary 2.2 (Refutation of Conjecture 5.1)**.** *For every fixed $h \in (0, 1]$, $N(c_{\mathrm{crit}}(N, h) - 1) \to 0$. Consequently, Conjecture 5.1 of [2] is false.*

Proof. By Theorem 2.1,

$$N(c_{\mathrm{crit}}(N, h) - 1) = \frac{1 + o(1)}{h\sqrt{\log N}} \longrightarrow 0,$$

which is incompatible with a positive $\Omega(N^{-1})$ lower bound for $c_{\mathrm{crit}}(N, h) - 1$. □

Henceforth we normalize $L = D = 1$, $x_\star = 0$, and $f(x_\star) = 0$. The general result follows from the transformation $f(x) = f(x_\star) + LD^2\widetilde{f}((x - x_\star)/D)$. The worst-case value is unchanged if the radius constraint is imposed with equality: for an instance of radius $r < D$, extend f to $\mathbb{R}^{d+1}$ by making it independent of the last coordinate and take that initial coordinate to be $\sqrt{D^2 - r^2}$. The trajectory and output gradients in the original coordinates are unchanged.

Lemma 2.3 (Huber lower bound)**.** *For every N, h, c in the stated ranges, $P_{N,h}(c; 1, 1) \geq (1 + Nhc)^{-1}$.*

Proof. Put $\eta = (1+Nhc)^{-1}$ and take $x_0 = 1$ and

$$f_\eta(x) = \begin{cases} \frac{1}{2}x^2, & |x| \le \eta, \\ \eta|x| - \frac{1}{2}\eta^2, & |x| > \eta. \end{cases}$$

All gradients queried by the algorithm equal η, and $x_i = 1 - ih\eta$ for $0 \le i \le N$. Thus $y_c = \eta$ and $|f'_\eta(y_c)| = \eta$. □

The supremum in (2.2) is finite. Cocoercivity gives $\|x_i\| \le \|x_0\| \le 1$ for $0 < h \le 1$, whence $\|y_c\| \le 2c-1$ and $\|\nabla f(y_c)\| \le 2c-1$. Lemma 4.2 below gives equality on $[1, 1+\delta]$ for some $\delta > 0$. For $f(x) = x^2/2$ and $x_0 = 1$, the output gradient has magnitude $|1 - c[1-(1-h)^N]|$, which exceeds $(1+Nhc)^{-1}$ for all sufficiently large c. Thus the set in (2.3) is nonempty and bounded, and $1 < c_{\mathrm{crit}}(N,h) < \infty$.

Outline of the proof. Let $\mathcal{V}_{N,h} = \{c \ge 1 : P_{N,h}(c;1,1) = (1+Nhc)^{-1}\}$, so that $c_{\mathrm{crit}} = \sup \mathcal{V}_{N,h}$. For fixed h and $0 < \varepsilon < 1$, write

$$c_\pm = 1 + \frac{1 \pm \varepsilon}{hN\sqrt{\log N}}.$$

The proof establishes, for all sufficiently large N,

$$[1, c_-] \subseteq \mathcal{V}_{N,h} \subseteq [1, c_+). \tag{2.7}$$

The first inclusion follows from the interpolation certificate and energy lower bound in Sections 3–4. For the second, Section 5 realizes strictly feasible negative directions as planar smooth convex functions. Section 6 constructs such directions uniformly for

$$\frac{(1+\varepsilon)t}{\sqrt{\log N}} \le a \le 2t, \qquad a = N(c-1), \quad t = 1/h,$$

and uses a quadratic function for $a \ge 2t$. Taking the supremum in (2.7) then yields the theorem.

3. Interpolation certificates and an auxiliary quadratic program

In the normalized setting write $g_i = \nabla f(x_i)$ and $q = \nabla f(y_c)$. We index the queried gradients by $i = 0, \dots, N-1$, the output by o, and the minimizer by $\star$. In vector and graph notation, node N denotes the output o.

A nonnegative combination of interpolation inequalities cancels the function values and leaves a quadratic form in $g_i - q$. A positive semidefinite form then gives the gradient bound.

The smooth-convex interpolation inequalities are

$$I_{ij} := f_i - f_j - \langle g_j, x_i - x_j\rangle - \frac{1}{2}\|g_i - g_j\|^2 \ge 0 \qquad (i,j \in \{\star, 0, \dots, N-1, o\}). \tag{3.1}$$

Indeed, let $\varphi(x) = f(x) - \langle g_j, x\rangle$. Then x_j minimizes φ and the descent lemma gives

$$\varphi(x_j) \le \varphi(x_i - \nabla\varphi(x_i)) \le \varphi(x_i) - \tfrac{1}{2}\|g_i - g_j\|^2,$$

which is (3.1).

Set

$$S = 1 + Nhc, \qquad K_i = h(Nc - i) > 0, \qquad T = hc\mathbf{1}\mathbf{1}^\top - hL_< - I, \tag{3.2}$$

where $L_<$ is the strictly lower triangular all-ones matrix. Consider nonnegative weights F_{ij}, $0 \le i < j \le N$, satisfying

$$\sum_{j>i} F_{ij} - \sum_{k<i} F_{ki} = 1 \qquad (0 \le i < N). \tag{3.3}$$

Write e_i for the standard basis of $\mathbb{R}^N$, set $e_N = 0$, and denote by T_i the ith row of T. Define

$$\lambda_{ij} = \frac{2hcS}{K_i} F_{ij}, \qquad b_i = \sum_{j>i} \lambda_{ij} - \sum_{k<i} \lambda_{ki} - 2hc, \tag{3.4}$$

$$M(F) = h^2c^2\mathbf{1}\mathbf{1}^\top - hcS \sum_{i<j} \frac{F_{ij}}{K_i} [(e_i - e_j)T_i + T_i^\top (e_i - e_j)^\top]. \tag{3.5}$$

The coefficients b_i are positive, since

$$b_i = 2hc \left[\frac{S}{K_i} - 1 + S \sum_{k<i} F_{ki} \left(\frac{1}{K_i} - \frac{1}{K_k} \right) \right] > 0. \tag{3.6}$$

Lemma 3.1 (Forward-flow certificate)**.** *For every flow satisfying (3.3), with $v_i = g_i - q$, $v_N = 0$, and $z = y_c - q$, one has*

$$\begin{aligned} \|x_0\|^2 - S^2\|q\|^2 = \|z\|^2 + \sum_{i,j<N} M(F)_{ij} \langle v_i, v_j \rangle + 2hc \sum_{i<N} I_{\star i} + 2I_{\star o} + 2SI_{o\star} \\ + \sum_{i<j} \lambda_{ij} I_{ij} + \sum_{i<N} b_i I_{oi}. \end{aligned} \tag{3.7}$$

In particular, $M(F) \succeq 0$ implies $P_{N,h}(c; 1, 1) = (1 + Nhc)^{-1}$.

Proof. Lemma A.1 proves (3.7) by expansion. If $M(F) \succeq 0$, all terms on its right-hand side are nonnegative by (3.1), (3.6), and $F \geq 0$. Hence $S\|q\| \leq \|x_0\| \leq 1$, and Lemma 2.3 gives equality. □

The adjacent-chain flow $F_{i,i+1} = i + 1$ defines

$$M_0 = h^2c^2\mathbf{1}\mathbf{1}^\top - hcS(D^\top W T + T^\top W D), \quad W = \operatorname{diag}\left(\frac{i+1}{K_i} \right)_{i=0}^{N-1}, \tag{3.8}$$

where $(D\beta)_i = \beta_i - \beta_{i+1}$ and $\beta_N = 0$. Write $R = \operatorname{diag}(K_i^{-1})T$, $\tau = R\beta$, and $(A\beta)_i = \tau_i - \tau_{i+1}$ for $i < N - 1$. Let

$$\mathcal{K} = \{\beta \in \mathbb{R}^N : \mathbf{1}^\top \beta = 1,\ D\beta \geq 0,\ A\beta \geq 0\}, \qquad \Phi_{N,h}(c) = \min_{\beta \in \mathcal{K}} \beta^\top M_0 \beta. \tag{3.9}$$

Thus there are N monotonicity conditions on β (including $\beta_{N-1} \geq 0$), and $N - 1$ on τ. For $\beta \in \mathcal{K}$, write $\delta_i = \beta_i - \beta_{i+1}$ and

$$Q(\beta) = \beta^\top M_0 \beta = h^2c^2 - 2hcS \sum_{i=0}^{N-1} (i+1)\delta_i \tau_i. \tag{3.10}$$

The optimizer exists uniquely by Lemma 4.3. Appendix A uses its KKT multipliers to construct a flow satisfying $M(F)\beta = \Phi_{N,h}(c)\mathbf{1}$, and proves $M(F) \succeq 0$ when $\Phi_{N,h}(c) \geq 0$. This yields the following criterion.

Proposition 3.2 (Quadratic-program sufficiency)**.** *If $N \geq 2$, $0 < h < 1$, $Nh^2 \geq 1$, $c \geq 1$, and $\Phi_{N,h}(c) \geq 0$, then*

$$P_{N,h}(c; 1, 1) = \frac{1}{1 + Nhc}. \tag{3.11}$$

Proof. Proposition A.5 constructs a flow satisfying (3.3) and $M(F) \succeq 0$ from the optimizer of (3.9). Lemma 3.1 applies. □

4. Energy lower bound and exactness below the critical scale

Put

$$t = h^{-1} \geq 1, \quad a = N(c-1) \geq 0, \quad d_i = N + a - i, \quad \mathfrak{F} = N + a + t, \quad \zeta_i = \mathfrak{F} - i = d_i + t. \tag{4.1}$$

For $p \in \mathbb{R}^N$ with $p_N = 0$, define $w_i = \sum_{j=i}^{N-1} p_j$, $z_i = w_i/\zeta_i$, $z_N = 0$, and $u_i = z_i - z_{i+1}$ for $0 \leq i < N$. Set $u_N = 0$. The map $p \mapsto u$ is invertible, and

$$p_i = \zeta_i u_i + \sum_{j>i} u_j, \qquad \sum_i p_i = \mathfrak{F} \sum_i u_i, \qquad p_i - p_{i+1} = \zeta_i u_i - (\zeta_i - 2) u_{i+1}. \tag{4.2}$$

Set

$$\mathcal{E}_{a,t}(p) = \left(\sum_i p_i \right)^2 - 2\mathfrak{F} \sum_{i=0}^{N-1} (i+1)(p_i - p_{i+1}) \frac{w_i - t p_i}{d_i}. \tag{4.3}$$

Lemma 4.1 (Tridiagonal reduction)**.** *For every $a \geq 0, t \geq 1$,*

$$\mathcal{E}_{a,t}(p) = \mathfrak{F}^2 u^\top E u, \tag{4.4}$$

where E is symmetric tridiagonal with

$$E_{ii} = \frac{2t(i+1)\zeta_i(\zeta_i - 1)}{d_i \mathfrak{F}} - 1, \qquad E_{i,i+1} = -\frac{(t-1)(i+1)\zeta_i(\zeta_i - 2)}{d_i \mathfrak{F}}. \tag{4.5}$$

Moreover,

$$E \succeq \operatorname{diag}(e_0^0, \ldots, e_{N-1}^0) \succ 0, \qquad e_i^0 = 2i + 1 - \frac{2i(i+1)}{N+1}. \tag{4.6}$$

Proof. See Lemma B.1. □

If $\mathbf{1}^\top v = 0$, substitution into (3.8) gives

$$\frac{v^\top M_0 v}{h^2 c} = \mathcal{E}_{a,t}(v) > 0 \qquad (v \neq 0). \tag{4.7}$$

Lemma 4.2 (Initial interval of exactness)**.** *For every $N \geq 1$ and $h \in (0, 1]$, there exists $\delta > 0$ such that $P_{N,h}(c; 1, 1) = (1 + Nhc)^{-1}$ for all $c \in [1, 1 + \delta]$.*

Proof. At $c = 1$, substitution into (3.8) gives $v^\top M_0 v = h^2 \mathcal{E}_{0,1/h}(v)$ for every $v \in \mathbb{R}^N$. Lemma 4.1 therefore implies $M_0 \succ 0$ at $c = 1$. The entries of M_0 depend continuously on c, so $M_0 \succ 0$ on $[1, 1 + \delta]$ for some $\delta > 0$. The adjacent-chain flow satisfies (3.3), and Lemma 3.1 gives the result. □

Lemma 4.3 (Well-posedness of the auxiliary program)**.** *For $N \geq 1$, $0 < h \leq 1$, and $c \geq 1$, the set $\mathcal{K}$ is compact and contains a point satisfying both inequality constraints strictly. The objective is strictly convex on the affine hyperplane $\mathbf{1}^\top \beta = 1$, so it has a unique minimizer on $\mathcal{K}$, and the relative Slater condition holds.*

Proof. The set $\mathcal{K}$ is closed and contained in the simplex, hence compact. For the uniform vector $\bar{\beta}_i = 1/N$,

$$(R\bar{\beta})_i = \frac{1}{N} - \frac{1}{NK_i}, \qquad (A\bar{\beta})_i = \frac{h}{NK_iK_{i+1}} > 0 \quad (0 \leq i < N - 1).$$

For sufficiently small $\eta > 0$, the perturbation $\widehat{\beta}_i = N^{-1} + \eta((N-1)/2 - i)$ has sum one, positive last coordinate, and positive adjacent differences. By continuity, $A\widehat{\beta} > 0$ as well. For $N = 1$, the slope constraints are empty and $\widehat{\beta}_0 = 1$ is strictly feasible. Thus $\mathcal{K}$ satisfies the relative Slater condition. Compactness gives a minimizer, and (4.7) gives its uniqueness. □

Define the reference reciprocal sum

$$\mathcal{R}_N = \sum_{i=0}^{N-1} \frac{1}{e_i^0}. \tag{4.8}$$

From (4.6), (4.2), and weighted Cauchy–Schwarz,

$$\mathcal{E}_{a,t}(p) \geq \frac{1}{\mathcal{R}_N} \quad \text{whenever} \quad \sum_i p_i = 1. \tag{4.9}$$

Furthermore, $\mathcal{R}_N = \log N + O(1)$. Indeed, with $q_i = 2(i+1)(N-i)/(N+1)$,

$$e_i^0 = q_i - 1 + \frac{2(i+1)}{N+1}, \qquad \left|\frac{1}{e_i^0} - \frac{1}{q_i}\right| \leq 2\left(\frac{1}{(i+1)^2} + \frac{1}{(N-i)^2}\right),$$

while $\sum_i q_i^{-1} = \sum_{j=1}^{N} j^{-1}$.

For every $\beta \in \mathcal{K}$, one has

$$\beta_i \geq B := \frac{a}{N(a+t)}. \tag{4.10}$$

To see this, set $B_i = K_i(\beta_i - \tau_i)$; the slope constraints imply $B_0 \geq N^{-1}$ and $B_{i+1} \geq (1-h)(K_{i+1}/K_i)B_i \geq 0$, while $B_{N-1} = (1+ha)\beta_{N-1} - ha/N$. Now let $m = t/(a+t)$ and write

$$\beta = B\mathbf{1} + mp, \qquad p \geq 0, \quad \mathbf{1}^\top p = 1. \tag{4.11}$$

Writing $\Delta p_i = p_i - p_{i+1}$, the shifted slopes and differences are

$$\tau_i = B + m\frac{w_i - tp_i}{d_i}, \qquad \delta_i = m\Delta p_i + B\mathbf{1}_{\{i=N-1\}}.$$

Consequently, with $J = \sum_i (i+1)\Delta p_i(w_i - tp_i)/d_i$,

$$\sum_i (i+1)\delta_i\tau_i = B + m^2 J + \frac{NmB(1-t)}{a+1}p_{N-1}.$$

Substitution into (3.10), together with $c - m^2 - 2\mathfrak{F}B = -a^2/(a+t)^2 - a/N$, yields

$$\frac{Q(\beta)}{h^2 c} = m^2\mathcal{E}_{a,t}(p) - \frac{a^2}{(a+t)^2} - \frac{a}{N} + \frac{2\mathfrak{F}mBN(t-1)}{a+1}p_{N-1}. \tag{4.12}$$

The final term is nonnegative; hence

$$\frac{\Phi_{N,h}(c)}{h^2 c} \geq \frac{t^2/\mathcal{R}_N - a^2}{(a+t)^2} - \frac{a}{N}. \tag{4.13}$$

Proposition 4.4 (Exactness below the critical scale)**.** *Fix $h \in (0,1]$ and $0 < \varepsilon < 1$. For all sufficiently large N,*

$$P_{N,h}(c;1,1) = \frac{1}{1+Nhc} \quad \textit{for every } 1 \leq c \leq 1 + \frac{1-\varepsilon}{hN\sqrt{\log N}}.$$

Proof. Write $a = kt/\sqrt{\log N}$ with $0 \leq k \leq 1-\varepsilon$. The lower bound (4.13) gives, uniformly over these k,

$$\frac{\log N}{h^2 c}\Phi_{N,h}(c) \geq \frac{t^2 \log N/\mathcal{R}_N - k^2 t^2}{(a+t)^2} - \frac{a\log N}{N} = 1 - k^2 + o_{h,\varepsilon}(1).$$

Since $1 - k^2 \geq 1 - (1-\varepsilon)^2 > 0$, the right-hand side is positive for all sufficiently large N. If $h < 1$, we also have $Nh^2 \geq 1$ eventually, and Proposition 3.2 gives the asserted equality.

For $h = 1$, take N large enough for the same strict lower bound, and also $N > 1$. Fix any c in the indicated interval. The right-hand side of (4.13), with $a = N(c-1)$ fixed and $t = 1/h$,

is continuous at $h = 1$ and strictly positive there. Hence one can choose $h_m \uparrow 1$, $h_m < 1$, with $Nh_m^2 \geq 1$ and $\Phi_{N,h_m}(c) > 0$. Proposition 3.2 bounds the output gradient of every admissible instance by $(1 + Nh_m c)^{-1}$. For each fixed instance, its finite trajectory and output gradient are continuous in h. Passing to the limit gives the bound $(1 + Nc)^{-1}$ at $h = 1$. Taking the supremum and applying Lemma 2.3 gives equality. □

5. Negative directions and planar counterexamples

We perturb the constant-gradient Huber data while keeping the output gradient of norm one. A negative quadratic direction decreases the initial radius below $S = 1 + Nhc$, and a quartic correction enforces the adjacent interpolation inequalities. Convex conjugacy realizes these data as a smooth convex function; rescaling gives a counterexample of unit initial radius.

Proposition 5.1. *Let $N \geq 1$, $0 < h \leq 1$, and $c \geq 1$. Suppose $\beta \in \mathbb{R}^N$ satisfies*

$$\mathbf{1}^\top \beta = 1, \qquad \beta_0 > \cdots > \beta_{N-1} > 0, \qquad (R\beta)_0 > \cdots > (R\beta)_{N-1}, \qquad Q(\beta) < 0. \tag{5.1}$$

Then $P_{N,h}(c; 1, 1) > (1 + Nhc)^{-1}$, witnessed by a globally defined 1-smooth convex function on $\mathbb{R}^2$. The conclusion also holds if $\Phi_{N,h}(c) < 0$.

Proof. Write $\tau = R\beta$, $\delta_i = \beta_i - \beta_{i+1}$ and extend $\beta_N = 0$. Define arrays A_i, B_i backwards from $A_N = B_N = 0$ by

$$A_i = A_{i+1} - \delta_i \tau_i, \qquad P_i = (TA)_i, \qquad B_i = B_{i+1} + \frac{1 - (A_i - A_{i+1})P_i}{K_i}, \qquad D_i = (TB)_i. \tag{5.2}$$

The recursion first determines A and $P = TA$, then B and $D = TB$. For $\epsilon > 0$, prescribe points and gradients in $\mathbb{R}^2$ by

$$q = y_c = (1, 0), \quad g_i = (1 + A_i\epsilon^2 + B_i\epsilon^4,\ \beta_i\epsilon), \quad x_0 = q + hc\sum_{i<N} g_i, \quad x_i = x_0 - h\sum_{j<i} g_j. \tag{5.3}$$

Set $(x_o, g_o, f_o) = (q, q, 1/2)$, $(x_\star, g_\star, f_\star) = (0, 0, 0)$, and $f_i = \langle g_i, x_i\rangle - \|g_i\|^2/2$ at ordinary nodes. These points satisfy (2.1).

For these data, the interpolation slack (3.1) reduces to

$$I_{ij} = \langle g_i - g_j, x_i - g_i\rangle. \tag{5.4}$$

If the source is $\star$ or o, it vanishes. For an ordinary source i we have $x_i - g_i = (K_i + P_i\epsilon^2 + D_i\epsilon^4,\ K_i\tau_i\epsilon)$. With $\Delta A = A_i - A_j$, $\Delta B = B_i - B_j$, $\Delta\beta = \beta_i - \beta_j$ and zero arrays at $j = o$, expansion gives

$$\begin{aligned} I_{ij} = {} & K_i(\Delta A + \tau_i\Delta\beta)\epsilon^2 + (K_i\Delta B + P_i\Delta A)\epsilon^4 \\ & + (D_i\Delta A + P_i\Delta B)\epsilon^6 + D_i\Delta B\epsilon^8. \end{aligned} \tag{5.5}$$

For $j = i + 1$, the quadratic coefficient is zero and the quartic coefficient equals 1. For $j \geq i + 2$ the quadratic coefficient is

$$K_i\sum_{\ell=i}^{j-1}\delta_\ell(\tau_i - \tau_\ell) > 0;$$

for $j < i$ it is $K_i\sum_{\ell=j}^{i-1}\delta_\ell(\tau_\ell - \tau_i) > 0$. The remaining ordinary-to-minimizer slacks equal $K_i + O(\epsilon^2) > 0$. The diagonal slacks are zero. Since there are finitely many ordered pairs, a single sufficiently small $\epsilon > 0$ makes every remaining slack nonnegative.

Moreover $\sum_i A_i = -\sum_i (i + 1)\delta_i\tau_i$, so

$$\|x_0\|^2 - S^2 = Q(\beta)\epsilon^2 + O(\epsilon^4) < 0. \tag{5.6}$$

To realize the data, define

$$\psi(u) = \tfrac{1}{2}\|u\|^2 + \max_{j\in\{\star,0,\dots,N-1,o\}} \langle x_j - g_j, u - g_j\rangle, \qquad f = \psi^*. \tag{5.7}$$

The function ψ is closed, finite and 1-strongly convex. For each $x \in \mathbb{R}^2$, the function $u \mapsto \psi(u) - \langle x, u\rangle$ is coercive and strongly convex, so it has a unique minimizer $u(x)$, characterized by $x \in \partial\psi(u(x))$. Hence $f = \psi^*$ is finite and differentiable, with $\nabla f(x) = u(x)$. Strong monotonicity of $\partial\psi$ gives

$$\|u(x) - u(y)\|^2 \le \langle x - y, u(x) - u(y)\rangle \le \|x - y\|\,\|u(x) - u(y)\|,$$

so ∇f is 1-Lipschitz. Convexity follows from the definition of the conjugate. At $u = g_i$, the affine term indexed by j equals $\langle x_j - g_j, g_i - g_j\rangle = -I_{ji} \le 0$, while the term indexed by i is zero. Thus $\psi(g_i) = \|g_i\|^2/2$ and $x_i \in \partial\psi(g_i)$. Fenchel equality gives $f(x_i) = \langle x_i, g_i\rangle - \psi(g_i) = f_i$ and $\nabla f(x_i) = g_i$. In particular, $\nabla f(0) = 0$, so 0 minimizes f.

Let $R_* = \|x_0\|$. By (5.6), $0 < R_* < S$. Rescale to $\widetilde{f}(x) = f(R_* x)/R_*^2$ and initial point x_0/R_*. This preserves the step size and the output relation, and produces $\|\nabla\widetilde{f}(y_c/R_*)\| = 1/R_* > 1/S$.

Finally, if $\Phi_{N,h}(c) < 0$, mix the optimizer of (3.9) with a strictly feasible point supplied by Lemma 4.3. A sufficiently small mixing weight preserves negativity of the objective and makes both orders strict, so the preceding construction applies. □

6. Uniform failure on the entire right half-axis

Fix $t = 1/h \ge 1$, put $s = 2t$, and define the trial profile

$$q_j^{\mathrm{H}} = \sum_{k=1}^{j} \frac{1}{k+s}, \qquad v_i = \frac{1}{i+s} + \frac{q_{N-i}^{\mathrm{H}}}{N}, \qquad H_N = \sum_{i=0}^{N-1} v_i, \qquad p_i = \frac{v_i}{H_N}. \tag{6.1}$$

We show that $\beta = B\mathbf{1} + mp$ is strictly feasible for $a \ge 0$ and has negative objective on $[(1+\varepsilon)t/\sqrt{\log N}, 2t]$ for large N. The quadratic example will cover $a \ge 2t$.

Lemma 6.1 (Strict feasibility for all nonnegative parameters)**.** *If $N \ge 4t$, then $\beta = B\mathbf{1} + mp$ is strictly feasible for every $a \ge 0$.*

Proof. The coordinates of v are strictly decreasing and positive; $m > 0$ and $B \ge 0$ therefore give $D\beta > 0$. For $i < N - 1$, put $j = N - i$ and use the slope functional (B.1). Its parameter dependence is affine:

$$\mathscr{L}_i(v; a) = \mathscr{L}_i(v; 0) + aC_i, \qquad C_i = tv_{i+1} - (t-1)v_i. \tag{6.2}$$

Lemma B.2, applied at $a = 0$, gives $\mathscr{L}_i(v; 0) > 0$. The definition of the profile gives

$$C_i = \frac{i + t + 1}{(i + 2t)(i + 2t + 1)} + \frac{1}{N}\left(q_{j-1}^{\mathrm{H}} - \frac{t-1}{j+2t}\right). \tag{6.3}$$

The second term is strictly greater than $-1/(2N)$, since $q_{j-1}^{\mathrm{H}} \ge 0$ and $(t-1)/(j+2t) < 1/2$. To bound the first, write $u = i + 2t$. It equals $(1 - t/(u+1))/u$. If $i < 2t$, then $u < 4t$ and this expression is greater than $1/(2u) > 1/(8t) \ge 1/(2N)$. If $i \ge 2t$, then $u \ge 4t$, $u < 3N/2$, and the expression is greater than $3/(4u) > 1/(2N)$. Thus $C_i > 0$. Equation (6.2) gives $\mathscr{L}_i(v; a) > 0$ for all $a \ge 0$, and the shifted slope identity at the end of Lemma B.2 gives $A\beta > 0$. □

Lemma 6.2 (Uniform compact-parameter upper energy bound)**.** *Fix $t \ge 1$ and $A \ge 0$. With the profile constructed using this fixed t,*

$$\sup_{0\le a\le A} \log N\, \mathcal{E}_{a,t}(p) \le 1 + o_{A,t}(1). \tag{6.4}$$

Proof. Set $v_N = 0$, $\gamma_i = (i+1)(v_i - v_{i+1}) \geq 0$, and $W_i = \sum_{k=i}^{N-1} v_k$. Lemma B.4 gives

$$N \sum_{i=0}^{N-1} \frac{\gamma_i v_i}{N-i} = O_t(1). \tag{6.5}$$

The exact difference (B.11) and $0 \leq W_i - v_i \leq (N-i-1)v_i$ imply

$$|J_{a,t}(v) - J_{0,1}(v)| \leq (a+t-1) \sum_i \frac{\gamma_i v_i}{N-i} = O_{A,t}(N^{-1})$$

uniformly for $a \in [0, A]$. Since $J_{0,1}(v) \geq 0$,

$$\begin{aligned} \mathcal{E}_{a,t}(v) - \mathcal{E}_{0,1}(v) &= -2(a+t-1)J_{0,1}(v) \\ &\quad - 2(N+a+t)\big[J_{a,t}(v) - J_{0,1}(v)\big] \leq O_{A,t}(1). \end{aligned}$$

Finally, Lemma B.3 gives $\mathcal{E}_{0,1}(v) = 4\log N + O_t(1)$, and (B.10) gives $H_N = 2\log N + O_t(1)$. Dividing by H_N^2 proves (6.4). □

Lemma 6.3 (Strict failure throughout the compact range)**.** *Fix $h \in (0,1]$ and $\varepsilon > 0$, and put $t = 1/h$. For all sufficiently large N, the trial objective is negative for every*

$$\frac{(1+\varepsilon)t}{\sqrt{\log N}} \leq a \leq 2t.$$

Consequently $P_{N,h}(1 + a/N; 1, 1) > (1 + Nh(1 + a/N))^{-1}$ throughout this range.

Proof. Multiplying (4.12) by $(a+t)^2$ gives the exact identity

$$\frac{(a+t)^2}{h^2 c} Q(\beta) = t^2 \mathcal{E}_{a,t}(p) - a^2 - \frac{a(a+t)^2}{N} + \mathcal{T}_{N,a,t}, \tag{6.6}$$

where the terminal term is

$$\mathcal{T}_{N,a,t} = \frac{2(N+a+t)ta(t-1)}{a+1}\, p_{N-1}.$$

Eventually $H_N \geq \log N$. Moreover

$$v_{N-1} = \frac{1}{N-1+2t} + \frac{1}{N(1+2t)} \leq \frac{2}{N}.$$

Thus $0 \leq \log N\, p_{N-1} \leq 2/N$. If $N \geq 4t$ and $0 \leq a \leq 2t$, then $N + a + t \leq 2N$, so

$$0 \leq \log N\, \mathcal{T}_{N,a,t} \leq 8t(t-1)a. \tag{6.7}$$

By Lemma 6.2 with $A = 2t$, eventually $\log N\, \mathcal{E}_{a,t}(p) < 1 + \varepsilon$ for every $a \in [0, 2t]$. Take N large enough that also

$$8t(t-1) \leq \frac{\varepsilon t}{2}\sqrt{\log N}.$$

Write $x = a\sqrt{\log N} \geq (1+\varepsilon)t$. Equations (6.6)–(6.7), after discarding the nonpositive cubic term, give

$$\begin{aligned} \log N\, \frac{(a+t)^2}{h^2 c} Q(\beta) &< t^2(1+\varepsilon) - x^2 + \frac{\varepsilon t}{2} x \\ &\leq t^2(1+\varepsilon) - (1+\varepsilon/2)tx \\ &\leq -\frac{\varepsilon}{2}(1+\varepsilon)t^2 < 0. \end{aligned}$$

Thus $Q(\beta) < 0$. Lemma 6.1 and Proposition 5.1 give the asserted planar examples. □

Lemma 6.4 (A quadratic covers the unbounded tail)**.** *Fix $h \in (0,1]$ and put $t = 1/h$. For all sufficiently large N and every $a \geq 2t$, the function $f(x) = x^2/2$ with $x_0 = 1$ has output gradient norm strictly greater than $(1 + Nh(1 + a/N))^{-1}$.*

Proof. Let $r_N = (1-h)^N$. Geometric decay gives $(N+2t)r_N \to 0$, including $r_N = 0$ when $h = 1$ and $N \geq 1$. For all sufficiently large N, $(N+2t)r_N < t$. For $c = 1 + a/N$, the output is $y_c = r_N - (a/N)(1 - r_N)$ and

$$N(-y_c) = a(1 - r_N) - Nr_N \geq 2t - (N + 2t)r_N > t.$$

Therefore

$$|f'(y_c)| = |y_c| \geq -y_c > \frac{t}{N} > \frac{t}{N + a + t} = \frac{1}{1 + Nhc}$$

for every $a \geq 2t$. □

Proposition 6.5 (Uniform exclusion of the entire right half-axis)**.** *For fixed $h \in (0,1]$ and $\varepsilon > 0$, all sufficiently large N satisfy (2.6) for every $c \geq c_+$ and every $L, D > 0$.*

Proof. For large N, the intervals $[(1+\varepsilon)t/\sqrt{\log N}, 2t]$ and $[2t, \infty)$ cover all $a = N(c-1)$ with $c \geq c_+$. Lemmas 6.3 and 6.4 give strict failure on these intervals, respectively. Both constructions have unit initial radius and dimension at most two. Scaling gives the result for $L, D > 0$. □

7. Proof of the main theorem

Proof of Theorem 2.1. Fix $h \in (0,1]$ and $0 < \varepsilon < 1$. In the normalized setting, Propositions 4.4 and 6.5 give

$$[1, c_-] \subseteq \mathcal{V}_{N,h} \subseteq [1, c_+)$$

for all sufficiently large N. Taking suprema yields

$$1 - \varepsilon \leq hN\sqrt{\log N}\,(c_{\mathrm{crit}}(N,h) - 1) \leq 1 + \varepsilon.$$

Letting $N \to \infty$ and then $\varepsilon \downarrow 0$ proves (2.4). Scaling restores arbitrary $L, D > 0$; the dimensional and radius assertions follow from the constructions in Proposition 6.5. □

Appendix A. Construction of the interpolation certificate

A.1. The forward-flow identity.

Lemma A.1. *The forward-flow identity (3.7) holds for any flow satisfying (3.3).*

Proof. Let $a_i = f_i - \langle g_i, x_i\rangle + \|g_i\|^2/2$. Then $I_{ij} = a_i - a_j + \langle g_i - g_j, x_i - g_i\rangle$. In the weighted interpolation sum on the right of (3.7), the coefficient of each ordinary a_i is $-2hc + \sum_{j>i}\lambda_{ij} - \sum_{k<i}\lambda_{ki} - b_i = 0$. Those of $a_o, a_\star$ vanish by (3.3) and $S = 1 + Nhc$. Furthermore,

$$x_0 = Sq + z + hc\sum_i v_i, \qquad x_i - g_i = K_i q + z + (Tv)_i.$$

Flow conservation implies $\sum_{i<j}\lambda_{ij}K_i(v_i - v_j) = 2hcS\sum_i v_i$. The weighted slacks therefore reduce to

$$2S\langle q, z\rangle + 2hc\langle\sum_i v_i, z\rangle + 2hcS\langle q, \sum_i v_i\rangle + \sum_{i<j}\lambda_{ij}\langle v_i - v_j, (Tv)_i\rangle.$$

Expanding the squared norms and using (3.5) gives the remaining quadratic term in (3.7). □

A.2. **The optimizer and its order structure.** Assume $N \geq 2$, $0 < h < 1$, and $c \geq 1$, and write $r = 1 - h$. By Lemma 4.3, the optimizer β is unique and admits KKT multipliers $\mu \in \mathbb{R}_+^{N-1}$ and $\nu \in \mathbb{R}_+^N$ satisfying

$$2M_0\beta = 2\Phi_{N,h}(c)\mathbf{1} + A^\top \mu + D^\top \nu, \qquad \mu_i(\tau_i - \tau_{i+1}) = 0, \quad \nu_i \delta_i = 0. \tag{A.1}$$

The equality multiplier is $2\Phi_{N,h}(c)$ by multiplication by $\beta^\top$. Define $B_i = K_i(\beta_i - \tau_i)$. A direct calculation gives

$$B_0 = S\beta_0 - hc \geq N^{-1}, \quad K_{i+1}(\tau_i - \tau_{i+1}) = \frac{hB_i}{K_i} - \delta_i, \quad B_{i+1} = B_i - (K_{i+1} + 1)\delta_i. \tag{A.2}$$

Thus $B_{i+1} \geq r(K_{i+1}/K_i)B_i > 0$ and $\beta_{N-1} > 0$.

Lemma A.2 (Strict order)**.** *For $0 < h < 1$, the optimizer satisfies $\beta_0 > \cdots > \beta_{N-1} > 0$. In particular, $\nu = 0$.*

Proof. Suppose a maximal constant block $\beta_l = \cdots = \beta_b$ has $l < b$. With the convention $\nu_{-1} = 0$, complementarity at the bounding differences gives $\nu_{l-1} = \nu_b = 0$. By (A.2), $B_i = B > 0$ on the block and $\tau_i = \beta_l - B/K_i$, hence $\tau_i - \tau_{i+1} > 0$ for $l \leq i < b$, and $\mu_i = 0$ there. Put

$$e_i = \frac{\mu_i}{2hcS}, \quad f_i = \frac{\nu_i}{2hcS}, \quad m_i = (i+1)\delta_i + e_i - e_{i-1}, \quad \omega_i = m_i/K_i,$$

with $e_{-1} = e_{N-1} = f_{-1} = f_N = 0$. Let $U_0 = \tau_0$ and $U_i = (i+1)\tau_i - i\tau_{i-1}$ for $i \geq 1$. To rewrite the KKT equations, observe that

$$D^\top W T \beta = U, \qquad \omega = WD\beta + \operatorname{diag}(K_i^{-1})(e_i - e_{i-1})_{i=0}^{N-1},$$

and

$$\frac{A^\top \mu}{2hcS} = T^\top \operatorname{diag}(K_i^{-1})(e_i - e_{i-1})_{i=0}^{N-1}.$$

Substituting (3.8) into (A.1) and dividing by $2hcS$ gives

$$U + T^\top \omega + D^\top f = \kappa \mathbf{1}, \qquad \kappa = \frac{hc}{S} - \frac{\Phi_{N,h}(c)}{hcS}. \tag{A.3}$$

For $\Omega = \sum_k \omega_k$,

$$(T^\top \omega)_i = hc\Omega - h\sum_{k>i} \omega_k - \omega_i, \qquad (D^\top f)_i = f_i - f_{i-1}.$$

Subtracting coordinate $i+1$ from coordinate i therefore gives

$$2f_i - f_{i-1} - f_{i+1} = \omega_i - r\omega_{i+1} - (U_i - U_{i+1}). \tag{A.4}$$

On the assumed constant block, $\omega_l \leq 0$, $\omega_i = 0$ for $l < i < b$, and $\omega_b > 0$, so the first two terms on the right have nonpositive sum. Meanwhile

$$\theta_i := \frac{i+1}{K_i} - \frac{i}{K_i + h} = \frac{h(Nc+1)}{K_i(K_i+h)}$$

is strictly increasing. For $l < i \leq b$, $U_i = \beta_l - B\theta_i$. At the left endpoint, $U_0 = \beta_0 - B\theta_0$ if $l = 0$, while for $l > 0$, (A.2) gives

$$\tau_{l-1} = \beta_l - \frac{B}{K_l + h} - \frac{r\delta_{l-1}}{K_l + h}, \qquad U_l = \beta_l - B\theta_l + \frac{lr\delta_{l-1}}{K_l + h} \geq \beta_l - B\theta_l.$$

Hence $U_i - U_{i+1} > 0$ for $l \leq i < b$. It follows that $2f_i - f_{i-1} - f_{i+1} < 0$ there, contradicting the existence of a maximum of a nonnegative sequence $f_l, \ldots, f_{b-1}$ with zero boundary values $f_{l-1} = f_b = 0$. □

By Lemma A.2, $f = 0$, so (A.3) reads $U + T^\top \omega = \kappa \mathbf{1}$. We retain $e_i \geq 0$, $e_{-1} = e_{N-1} = 0$, and $e_i(\tau_i - \tau_{i+1}) = 0$.

A.3. **Row moments and slope blocks.**

Lemma A.3 (Row moments and block endpoints)**.** *All m_i are positive and $\sum_i m_i = 1$. If $Nh^2 \geq 1$ and $\Phi_{N,h}(c) \geq 0$, every maximal constant block $[l, b]$ of τ satisfies $m_b \geq \delta_b$, with strict inequality at a terminal block.*

Proof. The sum of the m_i telescopes to $\sum_i (i+1)\delta_i = \sum_i \beta_i = 1$. On a maximal slope block $[l, b]$, put $g = \tau_{l-1} - \tau_l$ (zero for $l = 0$). Complementarity gives $e_{l-1} = e_b = 0$. If the block is a singleton, $m_l = (l+1)\delta_l > 0$. Otherwise the slope equality and KKT equations give

$$\delta_{i+1} = r\delta_i \ (l \leq i \leq b-2), \qquad \omega_i = r\omega_{i+1} \ (l < i < b), \qquad \omega_l = r\omega_{l+1} - lg. \tag{A.5}$$

Since $m_l = (l+1)\delta_l + e_l > 0$, the moment positivity propagates across the block.

For a nonterminal singleton, $m_b = (b+1)\delta_b \geq \delta_b$. For a nonterminal block of length at least two, the slope condition at its exit gives $\delta_b \leq r\delta_{b-1}$; hence $\delta_i \geq \delta_b$ for $l \leq i \leq b$. Summing m_i over the block, with $C = \omega_b$, yields

$$C \sum_{i=l}^{b} K_i r^{b-i} - K_l l g = \sum_{i=l}^{b} (i+1)\delta_i.$$

Using $K_i = K_b + h(b-i)$, $r^{b-i} \leq 1$, and $g \geq 0$ gives

$$\frac{m_b}{\delta_b} \geq \frac{K_b(l+b+2)}{2K_b + h(b-l)} \geq 1. \tag{A.6}$$

The last inequality follows from $K_b = h(Nc - b) \geq 2h$ for $b \leq N-2$, which implies $K_b(l+b) \geq h(b-l)$.

For the terminal block $b = N-1$, the singleton case gives $m_b = N\delta_b > \delta_b$. Otherwise write $B_* = \beta_b$, $C = \omega_b$, $\theta = \tau_b$, $\Omega = \sum_i \omega_i$. Since

$$\sum_i U_i = N\theta, \qquad \sum_i (T^\top \omega)_i = \sum_i (K_i - 1)\omega_i = 1 - \Omega,$$

summing (A.3) and then taking its last coordinate gives

$$N\kappa = N\theta + 1 - \Omega, \qquad \theta + hc\Omega - C = \kappa,$$

so $C = S\theta + \Phi_{N,h}(c)/(hc)$ and $\theta \leq C/S$. For $J = N - l$, let

$$D_J = \sum_{j=0}^{J-1} (K_b + hj) r^j, \qquad H_J = h \sum_{j=1}^{J-1} (N-j) r^{-j}.$$

On the constant slope block, (A.2) gives $\beta_{i+1} - \theta = r(\beta_i - \theta)$. Therefore

$$\delta_{b-j} = hr^{-j}(B_* - \theta) \quad (1 \leq j < J), \qquad \delta_b = B_*.$$

Summing the definition of m_i over the block cancels the e_i terms. Using (A.5) on the left and the preceding differences on the right gives

$$CD_J - K_l l g = \sum_{i=l}^{b} (i+1)\delta_i = NB_* + H_J(B_* - \theta).$$

Since $C > 0$, $g \geq 0$, and $\theta \leq C/S$, this implies

$$\frac{B_*}{C} \leq \frac{D_J + H_J/S}{N + H_J}. \tag{A.7}$$

We have $SK_b \geq (1+Nh)h > 1$, and the geometric series gives

$$D_J < \sum_{j=0}^{\infty}(K_b + hj)r^j = \frac{K_b + r}{h} \leq NK_b.$$

The last inequality follows from $(Nh-1)K_b - r \geq Nh^2 - 1 \geq 0$. Hence the right side of (A.7) is strictly less than K_b. This proves $m_b = K_b C > B_* = \delta_b$. □

A.4. **Construction of the directed flow.**

Lemma A.4 (Splitting probabilities and flow)**.** *Assume $N \geq 2$, $0 < h < 1$, $Nh^2 \geq 1$, $c \geq 1$, and $\Phi_{N,h}(c) \geq 0$. For the optimizer and multipliers above, set*

$$\ell_i = \delta_i + \frac{e_i}{i+1}, \quad \pi_0 = \pi_N = 1, \quad \pi_i = 1 - \frac{e_{i-1}}{i\ell_i}\ (1 \leq i < N), \quad s_i = 1 - \pi_i, \quad O_i = 1 + i\pi_i = \frac{m_i}{\ell_i}. \tag{A.8}$$

Then $0 \leq \pi_i \leq 1$, and

$$F_{ij} = O_i \pi_j \prod_{i<k<j} s_k \quad (0 \leq i < j \leq N) \tag{A.9}$$

is nonnegative, satisfies (3.3), and has row moments

$$\sum_{j>i} F_{ij}(\beta_i - \beta_j) = m_i \qquad (\beta_N = 0). \tag{A.10}$$

Every positive edge satisfies $\tau_i = \cdots = \tau_{j-1}$.

Proof. Since $e_{i-1} \geq 0$ and $\ell_i > 0$, $\pi_i \leq 1$. Put $G_0 = 0$ and $G_i = im_i - e_{i-1} = i(i+1)\ell_i\pi_i$ for $1 \leq i < N$. At the endpoints of a maximal slope block $[l, b]$, Lemma A.3 gives $G_l = lm_l \geq 0$ and $G_b = (b+1)(m_b - \delta_b) \geq 0$. On that block,

$$G_{i+1} - G_i = (i+1)(m_{i+1} - m_i + \delta_i).$$

Put $\chi_i = h(K_i - 1)\omega_{i+1} + \delta_i$ for $l \leq i < b$. By (A.5),

$$m_{i+1} - m_i + \delta_i = \chi_i + K_l lg\, \mathbf{1}_{\{i=l\}}.$$

For $K_i \geq 1$, $\chi_i > 0$. Once $K_i < 1$, define $R_i = \delta_i/[h(1-K_i)\omega_{i+1}]$; the sign of χ_i is the sign of $R_i - 1$. For successive indices in this range, (A.5) gives

$$\frac{R_{i+1}}{R_i} = r^2 \frac{1 - K_i}{1 - K_i + h} < 1.$$

Thus the sequence χ_i can change sign only from positive to negative. The additional nonnegative term affects only the first increment and preserves this property. Hence G_i first increases and then decreases on the block, so $G_i \geq \min\{G_l, G_b\} \geq 0$. This proves $\pi_i \geq 0$.

Since $\pi_j = 1 - s_j$ and $s_N = 0$, telescoping gives

$$\sum_{j=i+1}^{N} \pi_j \prod_{i<k<j} s_k = 1 - \prod_{k=i+1}^{N} s_k = 1.$$

Thus total outflow at i is O_i. To compute the inflow, put $V_i = \sum_{k<i} O_k \prod_{k<j<i} s_j$. Then $V_0 = 0$ and

$$V_{i+1} = O_i + s_i V_i.$$

If $V_i = i$, the right-hand side equals $1 + i\pi_i + is_i = i + 1$. Induction gives $V_i = i$, so total inflow is $i\pi_i$ and the divergence is $O_i - i\pi_i = 1$. If L_i denotes the mean gradient difference per unit outflow, then $L_{N-1} = \delta_{N-1}$ and $L_i = \delta_i + s_{i+1}L_{i+1}$. Since $s_{i+1}\ell_{i+1} = e_i/(i+1) = \ell_i - \delta_i$, backward induction gives $L_i = \ell_i$ and (A.10). Finally, if $\tau_{k-1} > \tau_k$, complementarity yields

$e_{k-1} = 0$, hence $s_k = 0$. For $i < k < j$, the product defining F_{ij} contains s_k, so $F_{ij} = 0$. Thus $F_{ij} > 0$ implies $\tau_i = \cdots = \tau_{j-1}$. □

A.5. **Positivity of the certificate matrix.**

Proposition A.5. *Under the hypotheses of Proposition 3.2, the flow (A.9) satisfies $M(F) \succeq 0$.*

Proof. Define a lower triangular matrix C by $C_{ii} = O_i/K_i$ and $C_{ji} = -F_{ij}/K_i$ for $i < j < N$. The row-moment identity gives $C^\top \beta = \omega$. The support property of the flow gives $CT\beta = U$; indeed, for $i \geq 1$,

$$(CT\beta)_i = O_i\tau_i - \sum_{k<i} F_{ki}\tau_k = (1 + i\pi_i)\tau_i - i\pi_i\tau_{i-1} = U_i,$$

where $(1 - \pi_i)(\tau_{i-1} - \tau_i) = 0$. The matrix satisfies $M(F) = h^2c^2\mathbf{1}\mathbf{1}^\top - hcS(CT + T^\top C^\top)$, so the KKT equation yields

$$M(F)\beta = \Phi_{N,h}(c)\mathbf{1}. \tag{A.11}$$

It remains to determine the signs of the off-diagonal entries. Put

$$\alpha_i = \frac{O_i}{K_i} = \frac{\omega_i}{\ell_i}, \quad Z_0 = 0, \quad Z_{i+1} = \alpha_i + s_i Z_i, \quad X_i = \frac{s_i Z_i}{\alpha_i}.$$

At the beginning of each maximal slope block, $s_i = X_i = 0$. Within a block,

$$X_{i+1} = \frac{\omega_i}{\omega_{i+1}}\left(1 - \frac{\delta_i}{\ell_i}\right)(1 + X_i) < r(1 + X_i),$$

and at a block boundary $X_{i+1} = 0$. Therefore $0 \leq X_i < r/h$ and $r\alpha_i - hs_iZ_i > 0$. Let $\rho_i = \alpha_i - \pi_i Z_i$. Flow conservation and $K_k > K_i$ for $k < i$ imply

$$\rho_i = \frac{\mathrm{out}_i}{K_i} - \sum_{k<i} \frac{F_{ki}}{K_k} \geq \frac{1}{K_i}.$$

To compute the off-diagonal entries, note that

$$Z_i = \sum_{k<i} \alpha_k \prod_{k<r<i} s_r.$$

For $i < j$, set $P_{ij} = \pi_j \prod_{i<k<j} s_k$. The product formula for the flow gives

$$\frac{F_{ij}}{K_i} = P_{ij}\alpha_i, \qquad \sum_{k\leq i} \frac{F_{kj}}{K_k} = P_{ij}(\alpha_i + s_iZ_i).$$

Using $T_{ki} = h(c-1) + h\mathbf{1}_{\{k\leq i\}} - \mathbf{1}_{\{k=i\}}$, we obtain

$$\begin{aligned}(CT)_{ij} &= hc\rho_i,\\ (CT)_{ji} &= h(c-1)\rho_j - h\sum_{k\leq i}\frac{F_{kj}}{K_k} + \frac{F_{ij}}{K_i}\\ &= h(c-1)\rho_j + P_{ij}(r\alpha_i - hs_iZ_i).\end{aligned}$$

Substitution into the expression for $M(F)$ gives

$$M(F)_{ij} = h^2c^2 - hcS\left[hc\rho_i + h(c-1)\rho_j + \pi_j \prod_{i<k<j} s_k\,(r\alpha_i - hs_iZ_i)\right] < 0, \tag{A.12}$$

since the bracket is at least $hc/K_i > hc/S$. We have $\beta_i > 0$ by Lemma A.2. Together with (A.11), this gives the exact decomposition, for every $z \in \mathbb{R}^N$,

$$z^\top M(F) z = \sum_{i<j} (-M(F)_{ij}) \beta_i \beta_j \left(\frac{z_i}{\beta_i} - \frac{z_j}{\beta_j} \right)^2 + \Phi_{N,h}(c) \sum_i \frac{z_i^2}{\beta_i} \geq 0. \tag{A.13}$$

Thus $M(F) \succeq 0$. □

Appendix B. Tridiagonal and endpoint estimates

B.1. Energy reduction and positivity.

Lemma B.1. *Equations (4.4)–(4.6) hold.*

Proof. By (4.2),

$$\frac{w_i - t p_i}{d_i} = z_i - \frac{t(\zeta_i - 1)}{d_i} u_i, \quad \sum_i (i+1)(p_i - p_{i+1}) z_i = \mathfrak{F} \sum_i u_i z_i - \sum_{i \geq 1} i(\zeta_i - 1) u_{i-1} u_i.$$

Using $2\sum_i u_i z_i = (\sum_i u_i)^2 + \sum_i u_i^2$, the squared sum cancels the first term of (4.3). The remaining diagonal and adjacent coefficients are (4.5), using $1 - t(\zeta_i - 1)/d_i = -(t-1)\zeta_i/d_i$ for the latter.

For positivity write

$$E = \operatorname{diag}(e_i(\mathfrak{F})) + \frac{t-1}{\mathfrak{F}} H, \qquad e_i(\mathfrak{F}) = 2i + 1 - \frac{2i(i+1)}{\mathfrak{F}},$$

where H is symmetric tridiagonal and

$$H_{ii} = \frac{2(i+1)\zeta_i^2}{d_i}, \qquad H_{i,i+1} = -\frac{(i+1)\zeta_i(\zeta_i - 2)}{d_i}.$$

Its off-diagonal entries are nonpositive. The sum of row i is at least

$$\frac{(i+1)\zeta_i(\zeta_i + 2)}{d_i} - \frac{i(\zeta_i^2 - 1)}{d_i + 1} > \frac{\zeta_i^2 + 2(i+1)\zeta_i + i}{d_i + 1} > 0.$$

At $i = N - 1$, the absent right off-diagonal entry contributes zero to the row sum. Writing the quadratic form as a sum of nonnegative weighted adjacent differences and strictly positive row-sum squares shows $H \succ 0$. Since $\mathfrak{F} \geq N + 1$ and $e_i(\mathfrak{F}) \geq e_i^0 > 0$, (4.6) follows. □

B.2. Strict feasibility of the trial vector.

Lemma B.2. *The vector $\beta = B\mathbf{1} + mp$ from (4.11) and (6.1) satisfies both strict orders for $0 \leq a \leq (2t+1)^{-1}$.*

Proof. Write $s = 2t$. The differences of v_i in (6.1) are positive, including $v_{N-1} > 0$; hence $D\beta > 0$. For $j = N - i \geq 2$, clearing the positive denominators in the slope inequality gives

$$\mathscr{L}_i(v) := (j + a + t) v_i - \sum_{k=i}^{N-1} v_k - t(j + a)(v_i - v_{i+1}) > 0. \tag{B.1}$$

We check the two summands of v separately. For $v_i^L = (i+s)^{-1}$, put $u = i + s \geq 2t$. Since

$$u \sum_{k=i}^{N-1} \frac{1}{k+s} = j - \sum_{k=0}^{j-1} \frac{k}{u+k}, \qquad \sum_{k=0}^{j-1} \frac{k}{u+k} \geq \frac{j(j-1)}{2(u+j-1)},$$

one obtains

$$u\mathscr{L}_i(v^L) \geq a\left(1-\frac{t}{u+1}\right)+t-\frac{tj}{u+1}+\frac{j(j-1)}{2(u+j-1)}>0.$$

The last three terms have numerator $2tu^2+(j-1)[j(u+1-2t)+2t]>0$ after multiplication by $2(u+1)(u+j-1)$; the first is nonnegative.

For $v_i^R=q_j^{\mathrm{H}}/N$, summation gives $\sum_{k=1}^{j}q_k^{\mathrm{H}}=(j+s+1)q_j^{\mathrm{H}}-j$, and hence

$$N\mathscr{L}_i(v^R)=j+(a-t-1)q_j^{\mathrm{H}}-\frac{t(j+a)}{j+2t}.$$

Using $q_j^{\mathrm{H}}\leq j/(2t+1)$ and $a\leq(2t+1)^{-1}$ gives

$$N\mathscr{L}_i(v^R)\geq\frac{t}{j+2t}\left(\frac{j(j-1)}{2t+1}-a\right)\geq\frac{t[j(j-1)-1]}{(j+2t)(2t+1)}>0.$$

Thus (B.1) holds. The shifted slope formula gives

$$\tau_i-\tau_{i+1}=\frac{m\,\mathscr{L}_i(v)}{H_N(j+a)(j+a-1)}>0\qquad(0\leq i<N-1),$$

so $A\beta>0$. □

B.3. **Reference energy and parameter estimates.** Comparison with a reciprocal model reduces the reference energy to two harmonic sums.

Lemma B.3 (Reference energy)**.** *For fixed $t\geq1$, the vector v in (6.1) satisfies*

$$\mathcal{E}_{0,1}(v)=4\log N+O_t(1).\tag{B.2}$$

Proof. Keep $s=2t$ fixed in v, and put $j=N-i$, $\xi_i=i+s$, $\Lambda=N+s$, and $L_i=\sum_{k=i}^{N-1}(k+s)^{-1}$. At energy parameters $(a,t)=(0,1)$, the transformed coordinate is

$$u_i^0=\frac{(j+1)/\xi_i-L_i+(j-sq_j^{\mathrm{H}})/N}{j(j+1)}.\tag{B.3}$$

Since the reference matrix is diagonal,

$$\mathcal{E}_{0,1}(v)=(N+1)^2\sum_i e_i^0(u_i^0)^2.\tag{B.4}$$

We compare u_i^0 with $\widehat{u}_i=(j\xi_i)^{-1}$. Write

$$r_i=L_i-\frac{j}{\Lambda}=\sum_{k=i}^{N-1}\frac{N-k}{\Lambda(k+s)}\geq0.$$

As $j+\xi_i=\Lambda$, subtraction in (B.3) gives

$$u_i^0-\widehat{u}_i=\frac{s}{N\Lambda(j+1)}-\frac{r_i}{j(j+1)}-\frac{sq_j^{\mathrm{H}}}{Nj(j+1)}.\tag{B.5}$$

Reversing the order of summation in the middle term gives

$$\begin{aligned}\sum_i\frac{r_i}{j(j+1)}&=\sum_{k=0}^{N-1}\frac{N-k}{\Lambda(k+s)}\sum_{i=0}^{k}\frac{1}{(N-i)(N-i+1)}\\&=\frac{1}{\Lambda(N+1)}\sum_{k=0}^{N-1}\frac{k+1}{k+s}\leq\frac{1}{N}.\end{aligned}$$

For the last term,

$$\sum_{j=1}^{N} \frac{q_j^{\mathrm{H}}}{j(j+1)} = \sum_{k=1}^{N} \frac{1}{k+s}\left(\frac{1}{k} - \frac{1}{N+1}\right) \le \sum_{k=1}^{N} \frac{1}{k(k+1)} \le 1.$$

The first term sums to at most s/N, since $\sum_{j=1}^{N}(j+1)^{-1} \le N$ and $\Lambda \ge N$. Consequently,

$$\sum_i |u_i^0 - \widehat{u}_i| \le \frac{1+2s}{N}. \tag{B.6}$$

To pass from this bound to the energy, note that $\xi_i^{-1} \le L_i \le j/\xi_i$ and $0 < j - sq_j^{\mathrm{H}} \le j$. Equation (B.3) therefore gives

$$0 < u_i^0 \le \widehat{u}_i + \frac{1}{N(j+1)}, \qquad u_i^0 + \widehat{u}_i \le (2+s)\widehat{u}_i,$$

where the second inequality uses $\xi_i \le Ns$. Also, $\xi_i j \ge N$ and $e_i^0 = 1 + 2ij/(N+1)$ imply

$$e_i^0 \le \frac{3\xi_i j}{N}, \qquad (N+1)^2 e_i^0 (u_i^0 + \widehat{u}_i) \le 12(2+s)N. \tag{B.7}$$

Define $\mathcal{M}_N = (N+1)^2 \sum_i e_i^0 \widehat{u}_i^2$. Factoring the difference of squares and using (B.6) yields

$$|\mathcal{E}_{0,1}(v) - \mathcal{M}_N| \le 12(2+s)N \sum_i |u_i^0 - \widehat{u}_i| \le 12(2+s)(1+2s). \tag{B.8}$$

It remains to sum the model. Partial fractions give

$$\frac{(N+1)^2 e_i^0}{j^2 \xi_i^2} = \mathsf{A}_N \left(\frac{1}{\xi_i} + \frac{1}{j}\right) + \frac{\mathsf{B}_N}{\xi_i^2} + \frac{\mathsf{D}_N}{j^2}, \tag{B.9}$$

where

$$\mathsf{A}_N = \frac{2N(N+1)}{\Lambda^2} + \frac{2(N+1)^2}{\Lambda^3} = 2 + O_t(N^{-1}),$$
$$\mathsf{B}_N = \frac{(N+1)^2}{\Lambda^2} - \frac{2s(N+1)}{\Lambda} = O_t(1), \qquad \mathsf{D}_N = \frac{(N+1)^2}{\Lambda^2} = O_t(1).$$

Both reciprocal sums in (B.9) equal $\log N + O_t(1)$, while the squared reciprocal sums are bounded. Thus $\mathcal{M}_N = 4\log N + O_t(1)$, and (B.8) proves (B.2). □

Lemma B.4 (Normalization and weighted profile bound)**.** *For fixed $t \ge 1$, the profile in (6.1), with $v_N = 0$ and $\gamma_i = (i+1)(v_i - v_{i+1})$, satisfies*

$$H_N = 2\log N + O_t(1), \qquad N \sum_{i=0}^{N-1} \frac{\gamma_i v_i}{N-i} = O_t(1).$$

Proof. Interchanging summations yields

$$H_N = \sum_{i=0}^{N-1} \frac{1}{i+s} + \frac{N+s+1}{N} q_N^{\mathrm{H}} - 1 = 2\log N + O_t(1). \tag{B.10}$$

For the weighted sum, put $j = N - i$ and split at $i = N/2$. For $i \le N/2$, use

$$v_i \le C_t\left((i+1)^{-1} + \frac{\log N}{N}\right), \quad \gamma_i \le C_t\left((i+1)^{-1} + \frac{i+1}{N^2}\right), \quad N/j \le 2.$$

Multiplying these bounds and summing gives

$$N\sum_{i\le N/2}\frac{\gamma_i v_i}{j}\le C_t\left(\sum_{i\ge 0}\frac{1}{(i+1)^2}+\frac{\log^2 N}{N}+1\right)=O_t(1).$$

For $i>N/2$ and $j\ge 2$, use $v_i\le C_t(1+\log j)/N$ and $\gamma_i\le C_t(N^{-1}+j^{-1})$. Then

$$N\sum_{\substack{i>N/2\\ j\ge 2}}\frac{\gamma_i v_i}{j}\le C_t\sum_{j=2}^{N}\left(\frac{1+\log j}{Nj}+\frac{1+\log j}{j^2}\right)=O_t(1).$$

At $j=1$, $v_{N-1}=O_t(N^{-1})$ and $\gamma_{N-1}=Nv_{N-1}=O_t(1)$ give the same estimate. □

Lemma B.5. *For fixed $t\ge 1$, uniformly for $0\le a\le (2t+1)^{-1}$,*

$$\mathcal{E}_{a,t}(p)=\frac{1}{\log N}+o_t((\log N)^{-1}).$$

Proof. We compare $\mathcal{E}_{a,t}(v)$ with $\mathcal{E}_{0,1}(v)$, keeping the profile v fixed with $s=2t$.

Put $\gamma_i=(i+1)(v_i-v_{i+1})$, $W_i=\sum_{k=i}^{N-1}v_k$, and $J_{a,t}(v)=\sum_i\gamma_i(W_i-tv_i)/(N+a-i)$. Then

$$J_{a,t}(v)-J_{0,1}(v)=-\sum_i\gamma_i\left[\frac{a(W_i-v_i)}{j(j+a)}+\frac{(t-1)v_i}{j+a}\right],\qquad j=N-i. \tag{B.11}$$

Using $0\le W_i-v_i\le (j-1)v_i$ gives

$$|J_{a,t}(v)-J_{0,1}(v)|\le (a+t-1)\sum_i\frac{\gamma_i v_i}{j}=O_t(N^{-1}).$$

The weighted sum is bounded by Lemma B.4.

Since $H_N^2-2(N+1)J_{0,1}(v)=\mathcal{E}_{0,1}(v)$, we have $J_{0,1}(v)=O_t(\log^2 N/N)$. Therefore

$$\mathcal{E}_{a,t}(v)-\mathcal{E}_{0,1}(v)=-2(a+t-1)J_{0,1}(v)-2\mathfrak{F}[J_{a,t}(v)-J_{0,1}(v)]=O_t(1)+O_t(\log^2 N/N),$$

uniformly in $a\in[0,(2t+1)^{-1}]$. Dividing by H_N^2 and using (B.10), (B.2) establishes the asserted energy asymptotics. □

Appendix C. Local transition of the quadratic program

The lower energy bound and the trial profile also determine the local asymptotics of $\Phi_{N,h}$ at the critical scale.

Lemma C.1 (Matching admissible profile)**.** *For $0\le a\le (2t+1)^{-1}$, the vector $\beta=B\mathbf{1}+mp$ defined by (4.11) and (6.1) satisfies $D\beta>0$ and $A\beta>0$. For fixed $t\ge 1$, uniformly over this range of a,*

$$H_N=2\log N+O_t(1),\qquad \mathcal{E}_{a,t}(p)=\frac{1}{\log N}+o_t((\log N)^{-1}). \tag{C.1}$$

Proof. Apply Lemmas B.2, B.4, and B.5. □

Theorem C.2 (Uniform transition of the quadratic program)**.** *Fix $h\in(0,1]$ and $K>0$. For $N\ge 2$, let $c_N(k)=1+k/(hN\sqrt{\log N})$. Then, as $N\to\infty$,*

$$\sup_{0\le k\le K}\left|\frac{\log N}{h^2 c_N(k)}\Phi_{N,h}(c_N(k))-(1-k^2)\right|\longrightarrow 0. \tag{C.2}$$

The same limit holds with $\Phi_{N,h}$ replaced by the objective of the strictly feasible trial vector from Lemma C.1.

Proof. With $a = N(c_N(k) - 1) = kt/\sqrt{\log N}$, the lower estimate (4.13) and $\mathcal{R}_N = \log N + O(1)$ give

$$\frac{\log N}{h^2 c_N(k)} \Phi_{N,h}(c_N(k)) \geq \frac{t^2 \log N/\mathcal{R}_N - k^2 t^2}{(a+t)^2} - \frac{a \log N}{N} = 1 - k^2 + o_{t,K}(1),$$

uniformly for $k \in [0, K]$. For large N, all these a satisfy the condition of Lemma C.1. Inserting its trial vector into (4.12) yields

$$\frac{Q(\beta)}{h^2 c_N(k)} = \frac{t^2}{(a+t)^2} \left(\frac{1}{\log N} + o_{t,K}((\log N)^{-1}) \right) - \frac{a^2}{(a+t)^2} - \frac{a}{N} + O_{t,K}\left(\frac{a}{\log N} \right).$$

Here $p_{N-1} = O_t((N \log N)^{-1})$ controls the terminal contribution. Multiplication by $\log N$ gives $1 - k^2 + o_{t,K}(1)$ uniformly. Since $\Phi_{N,h} \leq Q(\beta)$, the two estimates prove the claim. □

Verification and code

Theorem 2.1 and the refutation of Conjecture 5.1 are formalized in Lean 4.26.0 using mathlib. The declaration `original_sharp_critical_scale` establishes the limit for the supremum defined in (2.3). The formal proof has no unresolved proof obligations; its axiom audit contains only `propext`, `Classical.choice`, and `Quot.sound`. The accompanying source distribution fixes the dependencies and provides proof correspondence and verification records. From `lean_formalization/`, the project can be checked with `lake build`, and the theorem audit with `lake env lean OriginalAudit.lean`.

Supplementary Python programs use exact rational arithmetic to check finite instances of the interpolation identities, certificates, and counterexample constructions. They can be run with `python scripts/verify.py` from the project root. These checks complement the mathematical and formal proofs.

AI assistance

AI tools, including OpenAI Codex and other AI assistants, were used to explore proof strategies, perform algebraic and numerical calculations, develop verification code, assist the Lean formalization, and revise the exposition. The author is responsible for the mathematical statements, proofs, and final text.

References

[1] Y. Drori and M. Teboulle, *Performance of first-order methods for smooth convex minimization: A novel approach*, Math. Program. **145** (2014), 451–482.

[2] A. Luner and B. Grimmer, *On averaging and extrapolation for gradient descent*, Math. Oper. Res. **51** (2026), no. 3, 2158–2188. DOI: 10.1287/moor.2024.0449. See also arXiv:2402.12493v4, Conjecture 5.1.

[3] A. B. Taylor, J. M. Hendrickx, and F. Glineur, *Smooth strongly convex interpolation and exact worst-case performance of first-order methods*, Math. Program. **161** (2017), 307–345.

[4] M. Teboulle and Y. Vaisbourd, *An elementary approach to tight worst case complexity analysis of gradient based methods*, Math. Program. **201** (2023), 63–96.